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Segmentation only uses sparse annotations: Unified weakly and semi-supervised learning in medical images

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ABSTRACT

Since segmentation labeling is usually time-consuming and annotating medical images requires professional expertise, it is laborious to obtain a large-scale, high-quality annotated segmentation dataset. We propose a novel weakly- and semi-supervised framework named SOUSA (Segmentation Only Uses Sparse Annotations), aiming at learning from a small set of sparse annotated data and a large amount of unlabeled data. The proposed framework contains a teacher model and a student model. The student model is weakly supervised by scribbles and a Geodesic distance map derived from scribbles. Meanwhile, a large amount of unlabeled data with various perturbations are fed to student and teacher models. The consistency of their output predictions is imposed by Mean Square Error (MSE) loss and a carefully designed Multi-angle Projection Reconstruction (MPR) loss. Extensive experiments are conducted to demonstrate the robustness and generalization ability of our proposed method. Results show that our method outperforms weakly- and semi-supervised state-of-the-art methods on multiple datasets. Furthermore, our method achieves a competitive performance with some fully supervised methods with dense annotation when the size of the dataset is limited.

1. Introduction

Semantic segmentation is one of the fundamental tasks in computer vision. The rapid development of deep neural networks allows many advanced segmentation methods to achieve promising progress in recent years. However, the performance of Fully Supervised Learning (FSL) is heavily reliant on the quality and amount of pixel-level annotations. In particular, the manual annotations for medical images require professional skills by experienced radiologists, which is both cost-intensive and time-consuming, resulting in the scarcity of labeled data in clinical scenarios.

To reduce the intensive workload in pixel-level dense annotation for radiologists, many efforts have been made on the Weakly Supervised Learning (WSL) area (Dai et al., 2015; Lin et al., 2016; Bearman et al., 2016; Papandreou et al., 2015a), which aims at using coarse-grained annotations instead of precise segmentation masks to increase the efficiency of manual labeling procedure. However, the information of sparse annotations is much less than the dense annotation, which leads to a lower segmentation performance compare with FSL models using datasets of similar size. For instance, image-level labels (Papandreou et al., 2015a) cannot provide the position information of the target. While the bounding box (Dai et al., 2015) may indicate rough positions of the target, but the pixels inside the bounding box may belong to the

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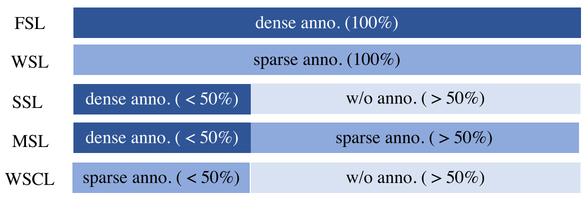

Fig. 1. Annotation settings for different supervision types.

background. Another labeler-friendly way to get the supervision efficiently is scribble-based annotation (Lin et al., 2016), which only requires the labeler to draw a few lines to mark a small part of the object or the background.

Another strategy to avoid the high cost of manual annotation is Semi-Supervised Learning (SSL) (Bai et al., 2017; Sedai et al., 2019; Bortsova et al., 2019), which trains networks jointly with a small number of labeled data and a large amount of unlabeled data. Extensive research on SSL shows that plenty of unlabeled data would benefit models' performance and generalization abilities. For example, Self-training methods (Bai et al., 2017; Laine and Aila, 2017; Tarvainen and Valpola, 2017) iteratively refine the pseudo-label for unlabeled data and learn from both labeled and pseudo-labeled data. Consistency regularization methods (Bortsova et al., 2019; Sajjadi et al., 2016; Bachman et al., 2014) encourage the model to predict consistent results when feeding input with different perturbations.

Several frameworks (Papandreou et al., 2015b; Li et al., 2018) have been designed in a hybrid way, Mixed-Supervised Learning (MSL). With only a small number of strongly (pixel-level) annotated images and a large number of weakly (bounding box or image-level) annotated images, the proposed methods could achieve comparable performance with the fully supervised system. The mixed-supervision setting still requires annotations, either strong or weak, for all images. In order to further reduce the cost of labeling while achieving comparable performance, we propose a framework following the setting that uses a small number of weakly labeled (scribble or points) data and a large amount of unlabeled data, called Weakly- and Semi-Supervised Consistency Learning (WSCL). Fig. 1 presents an overview of different supervision types.

In this paper, we propose an efficient and high-performance segmentation framework, named SOUSA, for WSCL, built upon the semi-supervised method following the mean teacher framework. The core idea can be summarized into two parts. (i) The main framework is a weakly supervised segmentation network with scribble annotations, and Geodesic distance map regression is introduced as an auxiliary task, which makes full use of sparse annotation and encourages the model to focus on the target area. (ii) We pass the unlabeled data to the student and mean teacher models with different perturbations. We then strengthen the consistency regularization for the predictions with a Mean Square Error loss and a carefully designed Multi-angle Projection Reconstruction (MPR) loss. In particular, MPR loss rotates the predictions with a random angle and then projects them along horizontal and vertical axes. Compared with the widely used MSE consistency loss, it penalizes the difference of the small false-positive region and amplifies the error on the boundary. We summarize our contributions in the paper as follows:

- We design a novel image segmentation framework SOUSA trained by a small amount of labeled data with scribble-based annotation and a large number of unlabeled data. To our best knowledge, this is the first work in WSCL setting on the medical image segmentation task.
- We propose a novel consistency loss function, i.e., MPR loss. It amplifies the prediction error, especially for the boundary and outliers, by projecting a segmentation map to uni-dimension.
- We show that our weakly- and semi-supervised learning method in the WSCL setting obtains significant improvement on the state-of-the-art weakly supervised methods and achieves a comparable performance of fully supervised models on a public dataset ACDC (Bernard et al., 2018) and an in-house colon cancer dataset.

2. Related works

2.1. Weakly-supervised semantic segmentation

To reduce the labor-intensive pixel-wise segmentation labeling, recent work has explored different forms of weak annotation for semantic segmentation, such as scribbles (Lin et al., 2016; Can et al., 2018; Tang et al., 2018; Lee and Jeong, 2020; Wu et al., 2021), bounding boxes (Khoreva et al., 2017; Dai et al., 2015; Zhao et al., 2018), points (Bearman et al., 2016; Laradji et al., 2021; Qu et al., 2019), image labels (Ahn and Kwak, 2018; Dubost et al., 2017), etc. This work focuses on scribble-based annotation, which is an intuitive and efficient substitution of segmentation labeling. During training, the scribbles of the background and each object are given, while the ground truth of other pixels remains unknown. The main challenges are the lack of supervision of object structure and the uncertainty on the boundary. Therefore, Lin et al. (2016) proposed *SribbleSup* to apply graph-based methods to propagate information to the unlabeled pixels. Can et al. (2018) suggested a two-step approach that first to train a CNN with scribbles and then to refine the CNN prediction with Conditional Random Fields (CRFs). Moreover, instead of using CRFs for post-processing, Tang et al. (2018) introduced a CRF-based regularization term that can be optimized together with the segmentation loss function.

Recently, Lee and Jeong (2020) introduced *Scribble2Label* architecture which leverages the consistency of predictions by iteratively averaging the predictions to improve the pseudo labels with label filtering. Valvano et al. (2021) proposed to generate segmentation masks in adversarial training with an attention gating mechanism which results in better object localization at multiple resolutions and achieves the state-of-the-art performance on the released scribble annotations for the ACDC dataset.

2.2. Semi-supervised semantic segmentation

Semi-Supervised Learning (SSL) (Chapelle et al., 2009) aims at leveraging a large amount of unlabeled data to improve the performance of the supervised model trained on a small dataset. Due to the scarcity of labeled medical image segmentation dataset, various SSL methods have been widely studied recently, including self-training (Bai et al., 2017; Laine and Aila, 2017; Tarvainen and Valpola, 2017), self-supervising (Doersch et al., 2015) and consistency regularization (Bortsova et al., 2019; Sajjadi et al., 2016; Bachman et al., 2014; Luo et al., 2021a; 2021b). Self-training models iteratively use pseudo-label predicted by the previously trained model as supervision for retraining. Various methods (Bai et al., 2017; Laine and Aila, 2017) are proposed to generate a better pseudo-label. FixMatch (Sohn et al., 2020) and PseudoSeg (Zou et al., 2021) use the pseudo segmentation of a weakly-perturbed image to supervise the prediction of a strongly perturbed image on a single network. Furthermore, Chen et al. (2021) design Cross Pseudo Supervision (CPS) utilizing two segmentation networks with different initialization for the same input image and using one network's prediction to supervise the other one and vice versa.

Consistency regularization methods leverage the unlabeled data with an intuitive assumption that a robust model should generate consistent predictions with similar inputs. To this end, the typical architecture (French et al., 2020) is to restrain the differences of the predictions with perturbed inputs, such as CutOut (Devries and Taylor, 2017), CutMix (Yun et al., 2019) or weak perturbation like Gaussian noise. Furthermore, Cui et al. (2019) adapt the mean teacher model so that the output of the teacher model could be more reliable and thus results in a better performance. Verma et al. (2019) adapt the conception of MixUp (Zhang et al., 2018) and propose Interpolation Consistency Training(ICT), which encourages the prediction for mixed-up inputs to be consistent with the mixed-up predictions for inputs.

2.3. Mixed supervised semantic segmentation

Mixed Supervised Learning (MSL) attempts to train networks with multiple types of annotations, usually combined with a small number of pixel-level annotated data and a large amount of weakly annotated data. For instance, Zhao et al. (2018) propose to utilize a mix of 3D bounding boxes for all instances while providing a voxel-wise label for only a tiny fraction of data. Mlynarski et al. (2018) extend segmentation networks with an additional branch predicting image-level labels for brain tumor segmentation tasks. Wang et al. (2020) introduces a label completion strategy to complete sparsely annotated data and then embed them into training data to enhance the model robustness. Moreover, Dolz et al. (2021) propose dual-branch architecture, where the teacher receives strong supervision while the student is supervised by weakly annotated data. Unlike MSL, we consider the WSCL setting in this paper, which trains a network with a small number of scribble-based annotations and a large number of unlabeled data.

3. Method

3.1. Problem setup and overview of framework

In this paper, as mentioned in Section 2.3, we consider a small dataset $\mathcal{D}_L$ with scribble annotation and a larger yet unlabeled dataset $\mathcal{D}_U$. The labeled dataset $\mathcal{D}_L = \{(x_i, y_i^s)\}_{i=1}^{N_L}$ contains N_L samples, where $x_i \in \mathbb{R}^{H\times W}$ is the ith input image with height of H and width of W and $y_i^s \in \mathbb{N}^{C\times H\times W}$ is the corresponding scribbles for each category including background. The unlabeled dataset $\mathcal{D}_U = \{u_j\}_{j=1}^{N_U}$ consists of N_U unlabeled images and N_U is usually much larger than N_L. In order to quantitatively evaluate different segmentation methods, the groundtruth masks y_i for labeled data are provided only in the evaluation stage.

The overview of proposed framework is illustrated in Fig. 2. The network f is adapted from the basic U-Net (Ronneberger et al., 2015) and modified following the concept of multi-task learning. It consists of two heads f_{seg} and f_{reg}, where $f_{seg} \in [0,1]^{C\times H\times W}$ is the segmentation head for pixel-wise segmentation and $f_{reg} \in [0,1]^{H\times W}$ is the regression head for estimating a distance map. For labeled data, we use the corresponding scribble and geodesic distance map generated from input image and scribble as supervisions to optimize the network with $\mathcal{L}_{seg}$ (Eq. 1) and $\mathcal{L}_{reg}$ (Eq. 3). Following the classic design of consistency-based Semi-supervised frameworks (Cui et al., 2019), we also maintain a momentum-updated teacher network, whose parameters are denoted as $\overline{\theta}$. The unlabeled images are fed into both $f(\cdot,\theta)$ and $f(\cdot,\overline{\theta})$ with different augmentations and we then constraint the consistency of the two segmentation predictions by similarity loss $\mathcal{L}_{mse}$ (Eq. 5) and the proposed novel consistency loss $\mathcal{L}_{mpr}$ (Eq. 8). We update the parameters of student network θ by gradient back-propagation while updating the parameters of teacher network $\overline{\theta}$ by exponential moving average of θ.

3.2. Segmentation supervised by scribbles

For the data whose scribbles are available, i.e., $\mathcal{D}_L = \{(x_i, y_i^s)\}_{i=1}^{N_L}$, we train the network with two tasks. The main task is to predict the segmentation masks, and the auxiliary task is to predict the distance map.

For the segmentation task, we minimize the pixel-wise segmentation loss between the prediction of segmentation head $f_{seg}(x_i,\theta)$ and the corresponding scribble y_i^s. As suggested in most scribble-based weakly supervised segmentation methods (Lee and Jeong, 2020; Valvano et al., 2021), we ignore the unlabeled pixels during the gradient back-propagation and apply the partially supervised Cross-Entropy loss. We can write the Partial Cross-Entropy (PCE) loss as a sampling of the loss with full masks:

$$\mathcal{L}_{seg}(f_{seg}(x,\theta), y^s) = -\sum_{c\in\mathcal{C}} y_c^s log(f_{seg}(x,\theta)_c) \tag{1}$$

where $f_{seg}(x,\theta)_c$ and y_c^s refer to the prediction and the corresponding scribble for category $c \in \mathcal{C}$ respectively.

One drawback for PCE is that most pixels in the region of interest but not in the scribble will be ignored even though they might be helpful for training. To this end, a common way is to generate a pseudo mask containing more labeled pixels with the help of input image and scribble, such as Graph-based methods (Ji et al., 2019) and GAN-based methods (Valvano et al., 2021). In this work, we propose to predict a pseudo mask based on a Geodesic distance map as an auxiliary task.

We first generate the pseudo mask offline for each category with Geodesic distance introduced in Wang et al. (2019), Luo et al. (2021d) and then merge them pixel-wisely. Compared with the Euclidean distance, the Geodesic distance also considers the intensity difference. Therefore, it is content-aware and provides more information for the region of interest (ROI) that the model should focus on. Visualization example for Geodesic distance map is shown in Fig. 3. For multiple categories, in our case, we first generate Geodesic distance maps for each category except the background, then combine them and keeping the pixel-wise max value.

$$y^r = \max_{c\in C} \mathcal{G}(x, y_c^s) \tag{2}$$

where $\mathcal{G}(x, y_c^s)$ represents the Geodesic distance map derived from input image x and its scribbles y_c^s. We modify the regression loss as follows.

$$\mathcal{L}_{reg}(f_{reg}(x,\theta), y^r) = \frac{1}{H\times W}\sum_{h=1}^{H}\sum_{w=1}^{W}[(f_{reg}(x,\theta) - y^r)^2]_{(h,w)} \tag{3}$$

The weakly supervised loss for labeled data is expressed as:

$$\mathcal{L}_{labeled} = \mathcal{L}_{seg} + \alpha\mathcal{L}_{reg} \tag{4}$$

where α is the weight balance parameter.

3.3. Consistency constraint with rotational maximal intensity projection

Inspired by Verma et al. (2019) and Cui et al. (2019), we design a mean-teacher framework. For images without annotation, we feed them into both student model and mean teacher model with different perturbation and then constraint the consistency between these outputs by pixel-wise Mean Squared Error (MSE) loss:

$$\begin{aligned}&\mathcal{L}_{mse}(f_{seg}(u,\theta), f_{seg}(\mathcal{T}(u),\overline{\theta}))\\ &= \tfrac{1}{H\times W}\textstyle\sum_{h=1}^{H}\sum_{w=1}^{W}[(f_{seg}(u,\theta) - f_{seg}(\mathcal{T}(u),\overline{\theta}))^2]_{(h,w)}\end{aligned} \tag{5}$$

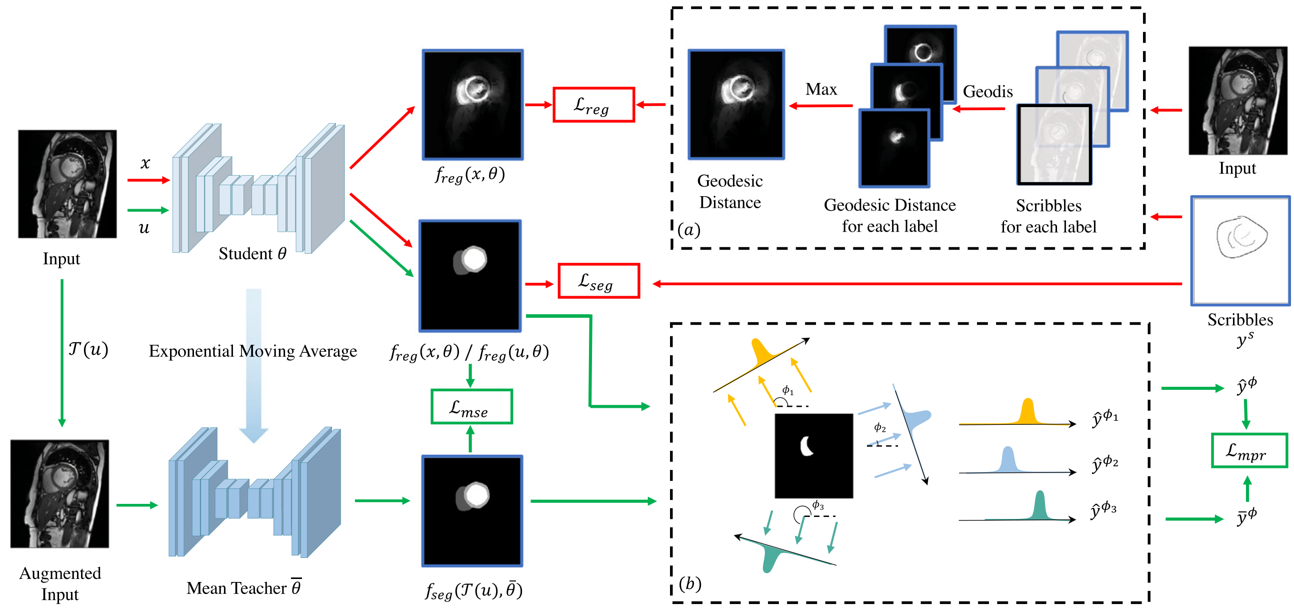

Fig. 2. An overview of proposed network architecture. (a) Procedure of generating the Geodesic Distance map with the input image and scribbles annotations. (b) An example of MPR loss.

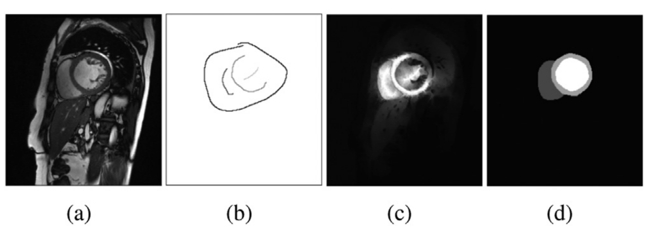

Fig. 3. The visualization of Geodesic distance map. (a) input image. (b) the corresponding scribbles, each gray-scale intensity represents one target. (c) Geodesic distance map. (d) Groundtruth mask.

where $\mathcal{T}(\cdot)$ represents intensity perturbation methods such as random noise and linear distortion, θ and $\bar{\theta}$ denote the parameters of student model and mean teacher model respectively.

Although MSE loss has been widely used in consistency-based semi-supervised learning, it suffers low sensitivity on small false-positive and boundary regions. To this end, we propose a novel consistency-based method named Multi-angle Projection Reconstruction (MPR) loss. As illustrated in Fig. 2(b), in addition to enhance the consistency of predictions of student model $\hat{y}_\theta \in [0,1]^{C\times H\times W}$ and mean teacher model $\hat{y}_{\bar{\theta}} \in [0,1]^{C\times H\times W}$, we randomly rotate them with same angle $\phi \in (0,\pi)$ and get $Rot_\phi(\hat{y}_\theta)$ and $Rot_\phi(\hat{y}_{\bar{\theta}})$. Then we project them along horizontal and vertical axes as follow.

$$Proj_{\phi,W}(\hat{y}_\theta) = \max_{1\le h\le H}[Rot_\phi(\hat{y}_\theta)]_{(h,\cdot)} \in [0,1]^{C\times H} \tag{6}$$

$$Proj_{\phi,H}(\hat{y}_\theta) = \max_{1\le w\le W}[Rot_\phi(\hat{y}_\theta)]_{(\cdot,w)} \in [0,1]^{C\times W} \tag{7}$$

where $Proj_{\phi,W}(\cdot)$ and $Proj_{\phi,H}(\cdot)$ are the horizontal and vertical projection vector respectively, and we note them as $Proj_\phi(\cdot)$ for convenience. We then constraint the consistency by MPR loss:

$$\mathcal{L}_{mpr}(\hat{y}_\theta,\hat{y}_{\bar{\theta}}) = 1 - \frac{1}{N_{rot}}\sum_{\phi\in\Phi}\frac{2\times Proj_\phi(\hat{y}_\theta)\times Proj_\phi(\hat{y}_{\bar{\theta}})}{Proj_\phi(\hat{y}_\theta)+Proj_\phi(\hat{y}_{\bar{\theta}})} \tag{8}$$

where $\Phi = [0,\pi]^{N_{rot}}$ and N_{rot} denotes the number of rotation angles in Φ.

Compared with conventional consistency losses, such as MSE loss and Dice loss, the proposed MPR loss puts more penalization when the disagreement areas are far from the main prediction regions. As the example shown in Fig. 4, where two predictions have a large overlap region and a small disagreement region. If this disagreement region moves along the axis from near to far, the Dice loss and MSE loss remain unchanged while our proposed MPR loss increases gradually. It proves that MPR loss has better sensitivity in the distant disagreement areas.

The loss function for unlabeled data consists of two components MSE loss and MPR loss with a weight parameter β.

$$\mathcal{L}_{unlabeled} = \mathcal{L}_{mse} + \beta\mathcal{L}_{mpr} \tag{9}$$

Thus the overall loss function can be formulated as:

$$\mathcal{L} = \mathcal{L}_{labeled} + \gamma(t)\mathcal{L}_{unlabeled} \tag{10}$$

where $\gamma(t)$ is the exponential time-dependent Gaussian warming up function controlling the balance between the supervised loss and the unsupervised consistency loss (Luo et al., 2021c; Yu et al., 2019). $\gamma(t)$ is defined as:

$$\gamma(t) = exp(-5\times(1-\frac{t}{T})^2) \tag{11}$$

where t represents the current iteration and T is the total number of iteration. The entire training pipeline is summarized in Algorithm 1

4. Experiments and results

4.1. Data and implementation details

Dataset We use Automated Cardiac Diagnosis Challenge (ACDC) dataset to validate our proposed framework. The ACDC dataset comprises 100 cine-MRI exams obtained by different temporal resolutions and magnetic strengths(1.5T and 3T). Each exam has two frames, i.e., phases end-diastolic (ED) and end-systolic (ES) cardiac phases. For each frame, a fine-grained pixel-wise annotation is provided for the right ventricle (RV), left ventricle (LV), and myocardium (MYO). The manually drew scribbles for RV, MYO, LV, and background based on the segmentation masks are provided by Valvano et al. (2021).

Pre-processing We select 2-dimensional slices which contain the targets from the original cine-MRIs and re-sampled them to the

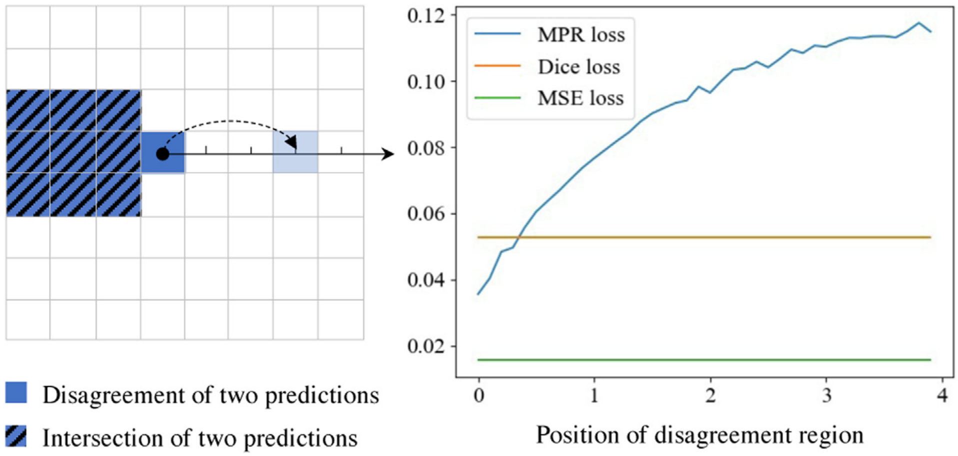

Fig. 4. A simplified example showing the advantage of MPR loss. Each grid contains 100 pixels, and N_{rot} is 360, i.e., randomly rotate 360 times. **Left:** The intersection and disagreement of two predictions. **Right:** We move the disagreement region along the axis from near to far and calculate the MPR loss, Dice loss and MSE loss for each position.

Algorithm 1 The overall algorithm of proposed method.

Require: N_{iter}: total number of iteration
Require: $\mathcal{D}_L$, $\mathcal{D}_U$: labeled and unlabeled dataset.
Require: f_{seg}, f_{reg}: two heads of proposed network.
Require: B: number of *labeled* samples each batch.
Require: U: number of *unlabeled* samples each batch.
Require: $\overline{\theta}_t$: parameters of *mean teacher* network.
Require: θ_t: parameters of *student* network.
Require: $\mathcal{T}(x)$: random perturbation on the input x.
Require: $\gamma(t)$: ramp function mentioned in Eq. 11.
Require: $\eta(t)$: learning rate decreasing w.r.t iteration.
Require: α, β: weights to balance multiple loss terms.
Require: δ: $\min\{1-1/(t+1), 0.99\}$ smoothing coefficient.
To initialize θ_0 and $\overline{\theta}_0$
$\theta_0 \leftarrow RandomInitialisation()$
$\overline{\theta}_0 \leftarrow \theta_0$
for $t = 0, \dots, N_{iter}$ **do**
  Sample $\{(x_i, y_i^s)\}_{i=1}^B \sim \mathcal{D}_L$
  # To Generate Geodesic distance map
  $\{y_i^r\}_{i=1}^B = \{Geodis(x_i, y_i^s)\}_{i=1}^B$
  # Weakly supervised loss
  $\mathcal{L}_{labeled} = \frac{1}{B}\sum_{i=1}^B(\mathcal{L}_{seg}(f_{seg}(x_i, \theta_t), y_i^s)$
    $+\alpha\mathcal{L}_{reg}(f_{reg}(x_i, \theta_t), y_i^r))$
  Sample $\{u_j\}_{j=1}^U \sim \mathcal{D}_U$
  for $\phi \in \Phi$ **do**
    # To project along horizontal and vertical axes
    $\{\hat{y}_{\theta,j}^{\phi}\}_{j=1}^U = \{Proj_\phi \circ f_{seg}(u_j, \theta_t)\}_{j=1}^U$
    $\{\hat{y}_{\overline{\theta},j}^{\phi}\}_{j=1}^U = \{Proj_\phi \circ f_{seg}(\mathcal{T}(u_j), \overline{\theta}_t)\}_{j=1}^U$
  end for
  # Semi-supervised loss
  $\mathcal{L}_{unlabeled} = \frac{1}{U}\sum_{i=1}^U(\mathcal{L}_{mse}(f_{seg}(u_j, \theta_t), f_{seg}(\mathcal{T}(u_j), \overline{\theta}))$
    $+\beta\sum_{\phi\in\Phi}\mathcal{L}_{mpr}(\hat{y}_{\theta,j}^{\phi}, \hat{y}_{\overline{\theta},j}^{\phi}\}))$
  $\mathcal{L} = \mathcal{L}_{labeled} + \gamma(t)\mathcal{L}_{unlabeled}$
  # To update parameters of student network
  $\theta_{t+1} \leftarrow \theta_t - \eta(t)\nabla_{\theta_t}\mathcal{L}$
  # To update parameters of mean teacher network
  $\overline{\theta}_{t+1} \leftarrow \delta\overline{\theta}_t + (1-\delta)\theta_t$
end for
return θ

same spacing of $1mm \times 1mm$. For normalization, intensity values in each slice were scaled to [0, 1] based on the minimal value and the 99-th percentile.

Implementation Details The dataset consists of 1902 slices from 100 patients. 5-fold cross-validation splitting by patient ID is employed for evaluation. Our segmentation framework is implemented in PyTorch on an NVIDIA 1080Ti GPU with 11GB memory. In this work, we use U-Net as the backbone for all experiments. A regression layer is added to the end of the U-Net for the distance map prediction. For each model, we randomly initialize the weights of the network and optimize them using the stochastic gradient descent(SGD) with momentum of 0.9 and weight decay of 10^{-4} for maximum 3×10^5 iterations. The learning rate is initialized as 0.01 and decreased every iteration. The batch size is set to 24. The hyperparameters setting is $\alpha = 5, \beta = 0.1$. The existing methods are implemented with the help of open source code base (Luo et al., 2020).

Metrics To quantitatively evaluate the performance of the proposed method, we measure the Dice Score (DSC), 95-th percentile Hausdorff Distance (HD) and Average symmetric surface distance (ASSD) in 3D volume. DSC measures volumetric overlap between the segmentation predictions and ground truths.

$$DSC = \frac{2 \times TP}{2 \times TP + FN + FP} \tag{12}$$

where TP, FP and FN are true positive, false positive and false negative, respectively. The Hausdorff distance is the maximum distance of a set to the nearest point in the other set, which can be defined as follow.

$$HD = max\{\max_{x\in X}\min_{y\in Y} d(x,y), \max_{y\in Y}\min_{x\in X} d(x,y)\} \tag{13}$$

where X, Y denote the surface points of two segmentation masks and $d(\cdot,\cdot)$ denotes the Euclidean distance between two points. Note that the metric HD mentioned in this paper is based on the calculation of the 95th percentile of the Hausdorff distances. ASSD measures the average symmetric surface distance between the segmented predictions and ground truths.

$$ASSD = \frac{1}{|X| + |Y|}(\sum_{x\in X}\min_{y\in Y} d(x,y) + \sum_{y\in Y}\min_{x\in X} d(x,y)) \tag{14}$$

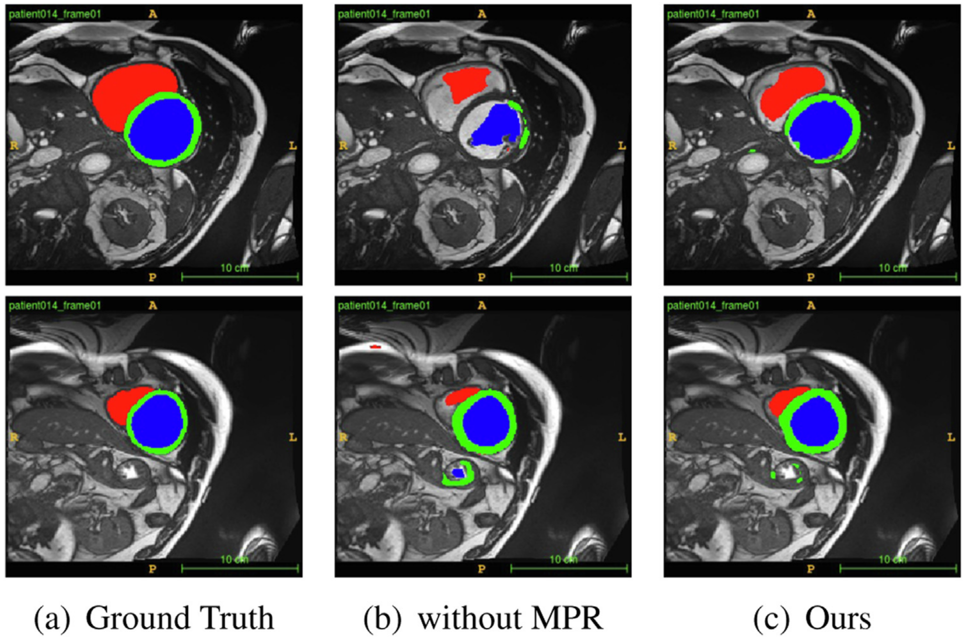

Fig. 5. Visualization of the improvement obtained by the MPR loss. *Best viewed in color.*

Table 1
Results of ablation study. 20% data is used. The best result for each column is marked in bold.

$\mathcal{L}_{seg}$	$\mathcal{L}_{mse}$	$\mathcal{L}_{reg}$	$\mathcal{L}_{mpr}$	$DSC_{(\%,\uparrow)}$	$HD_{(mm,\downarrow)}$	$ASSD_{(mm,\downarrow)}$
✓				$50.26_{\pm 3.90}$	$81.99_{\pm 5.88}$	$27.64_{\pm 2.07}$
✓	✓			$65.29_{\pm 4.51}$	$63.72_{\pm 6.64}$	$16.84_{\pm 4.37}$
✓	✓	✓		$79.35_{\pm 0.68}$	$20.16_{\pm 2.90}$	$5.95_{\pm 1.12}$
✓	✓	✓	✓	$\mathbf{81.69_{\pm 0.57}}$	$\mathbf{14.52_{\pm 3.73}}$	$\mathbf{3.96_{\pm 0.93}}$

Table 2
Results of ablation study. 20% data is used. The best result for each column is marked in bold.

N_{rot}	$DSC_{(\%,\uparrow)}$	$HD_{(mm,\downarrow)}$	$ASSD_{(mm,\downarrow)}$
1	$79.92_{\pm 0.99}$	$15.05_{\pm 4.13}$	$4.71_{\pm 1.33}$
3	$81.07_{\pm 0.55}$	$\mathbf{13.26_{\pm 3.27}}$	$\mathbf{3.84_{\pm 0.72}}$
9	$\mathbf{81.69_{\pm 0.57}}$	$14.52_{\pm 3.73}$	$3.96_{\pm 0.93}$
27	$79.19_{\pm 0.42}$	$14.78_{\pm 4.04}$	$4.03_{\pm 1.32}$

4.2. Ablation studies

We investigate the contribution of the three key components *i.e.*, semi-supervised learning, multi-task learning, and MPR. Table 1 shows the results.

The effect of semi-supervised learning $\mathcal{L}_{mse}$ The use of unlabeled data exhibits a significant performance gain. The consistency regularization, which aligns the probability map for unlabeled data, not only improves the accuracy of segmentation, DSC increase by 15.03% (65.29% vs. 50.26%) but also significantly reduces the false positive rate, HD and ASSD decrease by 18.27 mm (63.72 mm vs. 81.99 mm) and 10.80 mm (16.84 mm vs. 27.64 mm). It proves our hypothesis that utilizing a large amount of unlabeled data can improve the performance of weakly supervised learning.

The effect of regression loss $\mathcal{L}_{reg}$ The proposed multitask learning, which simultaneously predicts segmentation mask and attention map, improves the performance. Its DSC is higher than that of the network only predicts segmentation mask by 14.06%, while the HD and ASSD are much lower by 43.56 mm and 10.89 mm, respectively. We further calculate the metrics within the region, which is obtained by dilating the ground-truth mask by 5 mm. The DSC, HD, and ASSD of Multitask and without multitask are 77.75% vs. 76.90%, 8.43 mm vs. 9.05 mm, and 1.87 mm vs. 1.86 mm, respectively. The performance within the obtained region is similar, proving our hypothesis that the improvement of performance is caused by the attention map indicating the region of interest to reduce the false-positive rate.

The effect of MPR loss $\mathcal{L}_{mpr}$

The MPR loss is designed to emphasize the differences of the predictions from student and mean teacher, especially the boundary region and the outliers. The MPR loss further boosts the performance. The DSC increase by 2.34%. Fig. 5 shows two typical 2D slices that reflect the difference between ours and ours without MPR loss. We can observer that the MPR loss can reduce the small false positive region which is insensitive for mse consistency loss.

We compared the performance of the MPR loss with different rotation numbers, varying from 1 to 27, as shown in Table 2.

Effect of different weights of loss function We investigate the influence of the weight for different losses, i.e., the $\mathcal{L}_{reg}$ weight α and the $\mathcal{L}_{mpr}$ weight β. We set $\alpha = 5$ and $\beta = 0.1$ as the base setting. When evaluating one of the parameters, we fixed the other parameter. We evaluate our method with $\alpha \in \{1, 2, 5, 10, 20\}$, the results are given in Fig. 6(a). The best performance is achieved when $\alpha = 5$. When $\alpha = 20$, the performance suffers sharp degradation since the model focuses too much on the attention prediction, thus influencing the performance of segmentation in the multi-task scenario. We experiments with setting β as 0.01, 0.05, 0.1, 0.2, and 0.5. The results are shown in Fig. 6(b). Our method is robust to the hyperparameter β.

4.3. Comparison of different WSL and SSL methods

Our model is compared with models trained with pure sparse/dense annotation, several WSL and SSL methods, and we report the metrics of DSC, HD, and ASSD in Table 3. We further report the DSC of all methods on the left ventricle, right ventricle, and myocardium in Table 4.

Table 3
Results of Competing methods and ours in ACDC datasets.

Labeled ratio	10%			20%		
Method	DSC$_{(\%),\uparrow}$	HD$_{(mm,\downarrow)}$	ASSD$_{(mm,\downarrow)}$	DSC$_{(\%),\uparrow}$	HD$_{(mm,\downarrow)}$	ASSD$_{(mm,\downarrow)}$
Dense Anno. (Upper Bound)	70.76$_{\pm 1.75}$	14.25$_{\pm 0.80}$	3.69$_{\pm 0.23}$	78.94$_{\pm 0.67}$	9.68$_{\pm 0.69}$	1.80$_{\pm 0.04}$
Sparse Anno. (Lower Bound)	61.69$_{\pm 1.35}$	86.39$_{\pm 1.78}$	29.98$_{\pm 1.68}$	64.52$_{\pm 1.49}$	70.86$_{\pm 11.53}$	25.47$_{\pm 5.18}$
PCE+CRF (Tang et al., 2018)	70.93$_{\pm 1.69}$	11.84$_{\pm 1.65}$	2.82$_{\pm 0.54}$	77.17$_{\pm 0.41}$	8.84$_{\pm 0.31}$	2.05$_{\pm 0.22}$
S2L (Lee and Jeong, 2020)	67.62$_{\pm 2.49}$	53.57$_{\pm 4.84}$	15.86$_{\pm 2.41}$	70.74$_{\pm 0.39}$	53.35$_{\pm 1.23}$	14.53$_{\pm 0.57}$
MAAG (Valvano et al., 2021)	62.63$_{\pm 5.32}$	45.9$_{\pm 8.56}$	14.64$_{\pm 3.03}$	75.79$_{\pm 0.93}$	38.79$_{\pm 18.09}$	10.11$_{\pm 4.81}$
Self-training(Bai et al., 2017)	69.20$_{\pm 0.98}$	56.23$_{\pm 5.40}$	15.44$_{\pm 1.98}$	72.22$_{\pm 0.44}$	68.72$_{\pm 1.70}$	18.59$_{\pm 0.85}$
ICT (Verma et al., 2019)	72.46$_{\pm 0.39}$	42.61$_{\pm 3.08}$	11.34$_{\pm 0.56}$	73.96$_{\pm 0.52}$	45.37$_{\pm 3.12}$	11.76$_{\pm 0.77}$
Uncertainty (Yu et al., 2019)	66.41$_{\pm 3.86}$	34.14$_{\pm 0.78}$	9.51$_{\pm 3.38}$	74.94$_{\pm 0.30}$	21.84$_{\pm 3.28}$	5.73$_{\pm 1.01}$
Ours	76.01$_{\pm 0.35}$	16.41$_{\pm 1.71}$	4.46$_{\pm 0.61}$	79.54$_{\pm 0.50}$	12.07$_{\pm 1.42}$	3.26$_{\pm 0.56}$

Table 4
Dice scores of competing methods and our proposed method on LV, MYO, and RV.

Labeled ratio	10%			20%		
Method	LV$_{(\%),\uparrow}$	MYO$_{(\%),\uparrow}$	RV$_{(\%),\uparrow}$	LV$_{(\%),\uparrow}$	MYO$_{(\%),\uparrow}$	RV$_{(\%),\uparrow}$
Dense Anno. (Upper Bound)	70.46$_{\pm 6.13}$	79.10$_{\pm 3.04}$	86.74$_{\pm 2.57}$	80.74$_{\pm 3.16}$	84.03$_{\pm 1.16}$	90.21$_{\pm 1.98}$
Sparse Anno. (Lower Bound)	33.19$_{\pm 6.83}$	35.99$_{\pm 8.49}$	58.02$_{\pm 7.90}$	37.01$_{\pm 6.04}$	53.71$_{\pm 5.74}$	60.73$_{\pm 6.02}$
PCE+CRF Tang et al. (2018)	61.11$_{\pm 9.76}$	73.95$_{\pm 1.71}$	83.02$_{\pm 3.37}$	80.04$_{\pm 2.04}$	77.46$_{\pm 0.85}$	85.47$_{\pm 1.49}$
S2L Lee and Jeong (2020)	48.40$_{\pm 8.71}$	66.64$_{\pm 2.74}$	69.41$_{\pm 4.17}$	67.49$_{\pm 2.66}$	71.22$_{\pm 1.55}$	71.87$_{\pm 6.32}$
MAAG Valvano et al. (2021)	50.11$_{\pm 7.99}$	63.83$_{\pm 4.18}$	66.39$_{\pm 5.06}$	68.96$_{\pm 3.24}$	76.59$_{\pm 2.01}$	80.08$_{\pm 4.76}$
Self-training	63.93$_{\pm 5.15}$	64.73$_{\pm 4.76}$	72.73$_{\pm 6.93}$	72.99$_{\pm 5.80}$	70.28$_{\pm 4.30}$	79.60$_{\pm 4.26}$
ICT Verma et al. (2019)	68.21$_{\pm 5.02}$	65.61$_{\pm 4.20}$	74.73$_{\pm 9.28}$	76.70$_{\pm 3.72}$	72.80$_{\pm 3.21}$	81.97$_{\pm 3.59}$
Uncertainty Yu et al. (2019)	56.53$_{\pm 3.32}$	65.16$_{\pm 4.22}$	77.33$_{\pm 3.91}$	69.97$_{\pm 3.78}$	75.68$_{\pm 2.31}$	82.69$_{\pm 2.08}$
Ours	66.68$_{\pm 5.6}$	72.87$_{\pm 4.05}$	84.01$_{\pm 3.09}$	79.27$_{\pm 1.67}$	77.96$_{\pm 0.41}$	87.684$_{\pm 0.81}$

Fig. 6. The choice of hyperparameters. (a) α which balances the segmentation item and regularization item in weakly supervised loss function. (b) β which balances the MSE item and MPR item in semi-supervised loss function. The solid line corresponds to the mean dice scores. The light-colored band around the solid curve represents the 95% confidence level. *Best viewed in color.*

4.3.1. Compared with WSL methods

We compare our proposed method with WSL Methods such as scribble annotation(PCE), Regularized Loss (rLoss) (Tang et al., 2018) that integrates the DenseCRF into the loss function, and

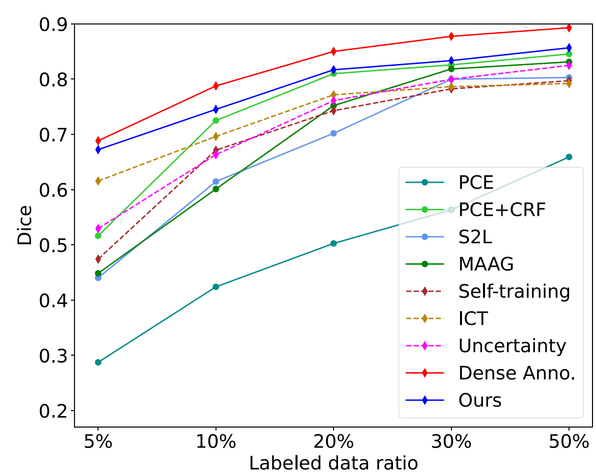

Fig. 7. The segmentation performance of our method and competing methods with different ratio of labeled data. *Best viewed in color.*

scribble2label (Lee and Jeong, 2020) that combines pseudo-labeling and label-filtering to generate more reliable labels.

The experiment results in Table 3 show that our method leads to the best DSC. Compared with PCE+CRF, the DSC increases by 5.08% (76.01% vs 70.93%) and 2.37% (79.54% vs 77.17%) with 10% and 20% labeled data, respectively. It should be noticed that the HD and ASSD of CRF are better than us. CRF considers the relationship between the current pixel and its neighboring pixels to reduce the small false positive regions. However, it can not make the network focus more on the foreground regions. Thus, it sometimes suffers under-segmentation in the foreground region, as shown in Fig. 8. As discussed in Section 1, our method is orthogonal to these methods. These methods can easily be applied to our method.

4.3.2. Compared with SSL methods

We also compared the proposed method to several semi-supervised methods. The self-training (Bai et al., 2017) used a pre-trained segmentation model trained with labeled data to generate pseudo-labels for unlabeled data. ICT (Verma et al., 2019) applied

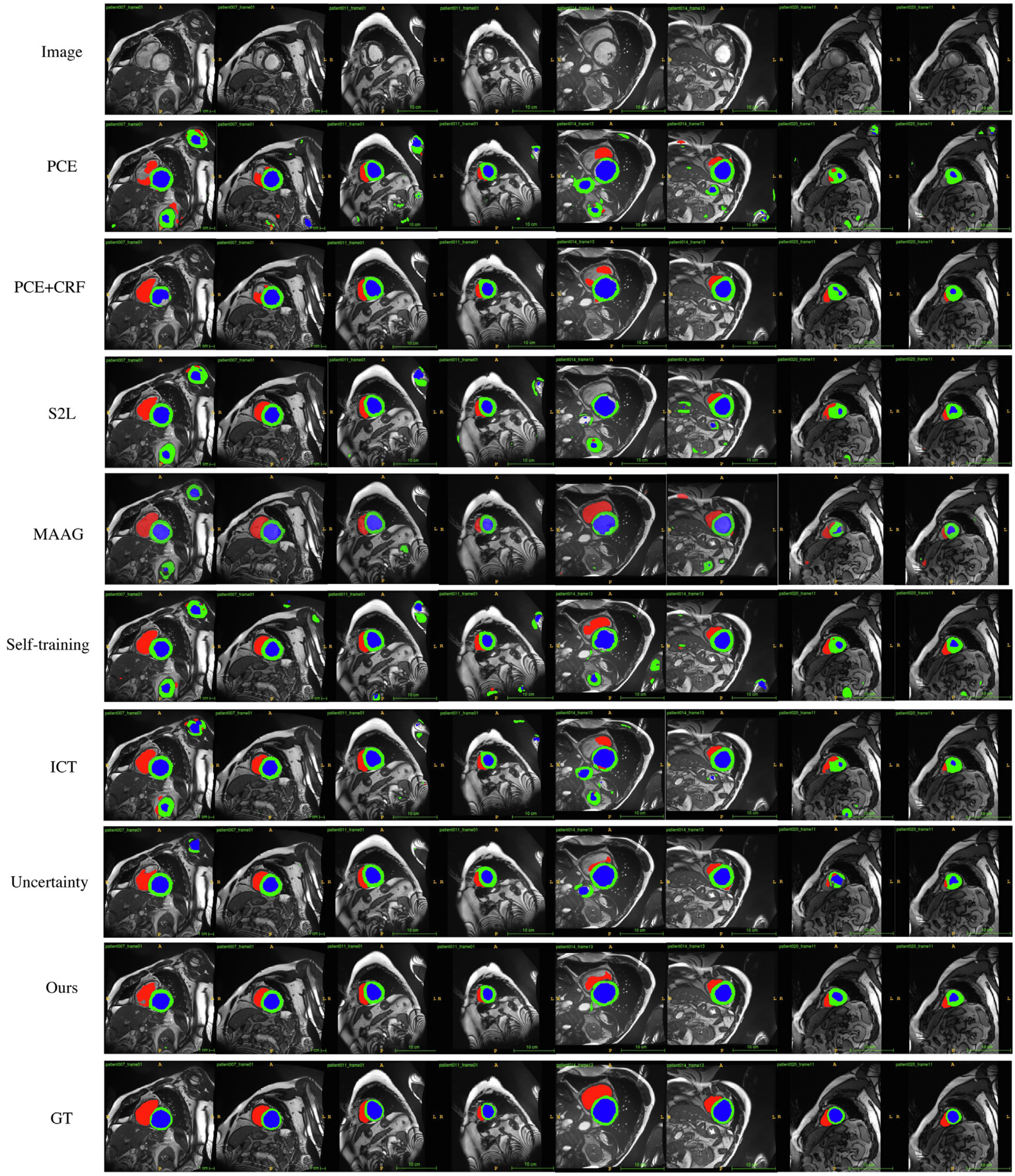

Fig. 8. Visualization of segmentation results obtained by different methods. *Best viewed in color.*

mixup as augmentation in the mean teacher framework. Uncertainty (Yu et al., 2019) exploited the uncertainty information to enable networks to learn from meaningful and reliable targets. CPS (Chen et al., 2021) utilized the predictions of two networks differently initialized to supervise each other.

Since we compare these methods in our WSCL scenario, the dense annotations are replaced by the sparse annotation, and PCE replaces the loss for the segmentation mask.

Compared with the ICT, the DSC increases by 3.55%, and the HD and ASSD decrease by 26.20 mm and 6.88 mm, respectively. As shown in Fig. 8, the false positive regions in our method are much less than those predicted by the other SSL methods. The results demonstrate that taking advantage of sparse annotations and applying powerful consistency realization could make our method performs better than those methods, especially in the HD and ASSD metrics.

4.3.3. *Effect of different ratio of labeled data*

We compared the segmentation performance of our method and other competing methods with different ratios of labeled data.

Table 5
Results of competing methods and ours in Colon datasets. The best result for each column is marked in bold. Let N_L and N_U denote the usage of labeled data and unlabeled data, respectively.

Method	$N_L : N_U$	$DSC_{(\%,\uparrow)}$	$HD_{(mm,\downarrow)}$	$ASSD_{(mm,\downarrow)}$
Sparse Anno.	50:0	$52.16_{\pm 3.18}$	$94.98_{\pm 6.46}$	$38.35_{\pm 5.34}$
Dense Anno.	50:0	$68.27_{\pm 0.72}$	$27.31_{\pm 1.59}$	$8.35_{\pm 0.53}$
Self-training (Bai et al., 2017)	50:50	$61.15_{\pm 1.47}$	$67.38_{\pm 8.40}$	$22.91_{\pm 2.60}$
	50:100	$62.30_{\pm 0.72}$	$55.38_{\pm 3.05}$	$18.87_{\pm 1.12}$
	50:150	$62.85_{\pm 1.20}$	$53.71_{\pm 2.91}$	$17.75_{\pm 1.23}$
	50:200	$64.20_{\pm 1.48}$	$59.72_{\pm 1.27}$	$18.89_{\pm 1.19}$
ICT (Verma et al., 2019)	50:50	$57.94_{\pm 0.41}$	$85.91_{\pm 2.69}$	$29.68_{\pm 1.59}$
	50:100	$60.15_{\pm 0.76}$	$71.64_{\pm 9.20}$	$23.75_{\pm 3.00}$
	50:150	$58.70_{\pm 1.09}$	$78.44_{\pm 4.55}$	$26.35_{\pm 1.17}$
	50:200	$57.96_{\pm 1.47}$	$84.86_{\pm 2.32}$	$29.03_{\pm 1.44}$
Uncertainty (Yu et al., 2019)	50:50	$60.25_{\pm 2.76}$	$95.20_{\pm 9.84}$	$27.31_{\pm 4.91}$
	50:100	$59.44_{\pm 4.72}$	$83.64_{\pm 26.71}$	$25.96_{\pm 9.04}$
	50:150	$61.43_{\pm 1.73}$	$59.64_{\pm 31.48}$	$18.33_{\pm 8.21}$
	50:200	$62.01_{\pm 3.96}$	$75.96_{\pm 36.59}$	$19.25_{\pm 8.68}$
Ours	50:50	$61.30_{\pm 0.97}$	$61.52_{\pm 2.96}$	$20.80_{\pm 1.82}$
	50:100	$62.48_{\pm 0.62}$	$65.55_{\pm 3.81}$	$21.16_{\pm 1.28}$
	50:150	$63.64_{\pm 1.26}$	$54.93_{\pm 2.06}$	$17.89_{\pm 0.70}$
	50:200	$\mathbf{66.38}_{\pm \mathbf{1.68}}$	$\mathbf{51.37}_{\pm \mathbf{3.97}}$	$\mathbf{15.11}_{\pm \mathbf{1.90}}$

The changes in DSC are shown in Fig. 7. It can be observed that our method performed best in all the data ratios compared to weakly supervised or semi-supervised methods. It can be found that less labeled data makes the gap between ours and other methods greater. When the labeled ratio is less than 5%, our method is comparable to the method trained with dense annotations.

Besides the WCSL scenario, the methods with 100% dense annotation (FSL) could achieve 90.49 of DSC, and the method with 100% sparse annotation (WSL) could achieve 68.73 of DSC, respectively.

4.4. Colon dataset

To further validate the proposed method, we apply it on a colon dataset collected from *The Sixth Affiliated Hospital of Sun Yat-sen University* and *The First People's Hospital of Kashi Prefecture* (Jin et al., 2021). The colon dataset consists of 275 abdominal CT exams from patients suffering from primary colon cancer. Acquisition parameters are: 120 kVp; exposure time, 800 ms; and X-Ray tube current, 350 mA. Reconstruction parameters are: slice thickness, 5.0 mm; and reconstruction diameter, 337–381 mm.

This study was approved by the Ethics Committee of The Sixth Affiliated Hospital of Sun Yat-sen University and The First People's Hospital of Kashi Prefecture. It was conducted in accordance with the most recent version of the Declaration of Helsinki.

The colon tumor segmentation masks are manually segmented using ITK Snap (Yushkevich et al., 2006) by expert radiologists. The scribbles are generated based on the masks following the same procedures mentioned in Valvano et al. (2021). We obtain the

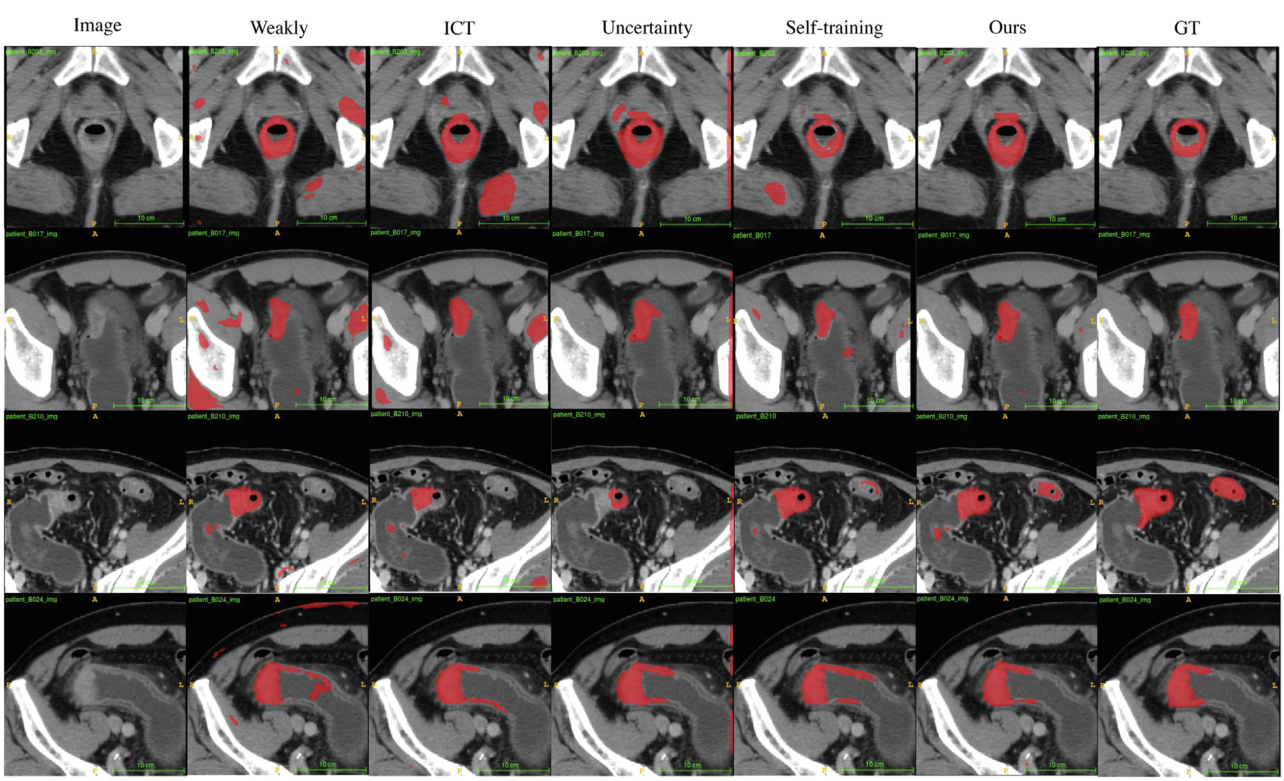

Fig. 9. Visualization of segmentation results obtained by different methods on colon dataset. *Best viewed in color.*

scribbles for foreground as well as background by standard skeletonization, which brings deterministic results for reproducibility.

As preprocessing, we first select the slices containing colon tumors and re-sample them to the spacing of $1mm \times 1mm$, then crop them to the size of $(256, 256)$ pixel and apply a normalization with fixed windowing $[-125, 275]$. The training set contains 923 labeled slices from 50 CT exams and 2670 unlabeled slices from 200 exams. The testing set is comprised of 417 slices from 25 CT scans. We run three times for each experiment and report the average and standard deviation of metrics. Colon dataset uses the same hyper-parameters as the ACDC dataset.

Compared with the ACDC dataset for organ (heart) segmentation, the Colon dataset is for tumor segmentation. For heart segmentation, the heart has a fixed shape, and the differences in each part are relatively large. But for tumor segmentation, the shape of the colon tumor and colon both change considerably, and the contrast of the colon image is very subtle. Hence, segmenting colon tumors is more complex than segmenting the heart.

We compare the performance of our proposed methods with state-of-the-art methods. Quantitative and qualitative results are shown in Table 5 and Fig. 9. The results show that semi-supervised methods leveraging a large number of unlabeled data indeed improve the performance of segmentation comparing to simple weakly supervised methods with only labeled data. In most cases, larger unlabeled datasets lead to better improvements. Comparing to state-of-the-art methods (ICT (Verma et al., 2019), Uncertainty aware (Yu et al., 2019) and Self-training), ours achieves the best performance in every data ratio. In particular, our method even achieves a comparable DSC (66.38%) with the method using dense annotation (68.27%) when 200 unlabeled data is used during semi-supervised training.

5. Conclusion

We introduce a novel framework SOUSA to learn semantic segmentation from data with scribble annotation and unlabeled data. In order to fully exploit the weak supervision, besides partial cross-entropy (PCE) loss, we design an auxiliary task to predict the Geodesic distance map of ROI. Following the semi-supervision concept, we propose imposing the consistency of predictions from student network and mean teacher network with MSE loss and MPR loss, which can reduce false-positive predictions.

Extensive experiments are conducted on various datasets to demonstrate that our method significantly outperforms both weakly supervised state-of-the-art methods (MAAG, Scribble2Label) and semi-supervised state-of-the-art methods (ICT, uncertainty-aware, Self-training). Furthermore, with sparse annotations and enough unlabeled data, our method achieves comparable performance with the fully supervised method with pixel-level annotation, effectively alleviating the intensive segmentation annotation workload. The improvement of segmentation accuracy and efficiency makes the proposed method highly valuable in real-world clinical scenarios where precise segmentation annotations are limited.

Declaration of Competing Interest

All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.

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